

Paper Checker – Team Member 2

Module: Knowledge Base + Evaluation Engine

Goal

Build the evaluation engine that compares a student's answer against NCERT concepts and generates meaningful feedback.

This module MUST NOT perform OCR.

Input:

Question (string)

Student Answer (string)

Output:

Evaluation Result

Folder Structure

backend/

└─ services/

└─ rag/

├─ textbook_loader.py

├─ retriever.py

├─ rubric.py

├─ prompts.py

└─ evaluation.py

data/

└─ ncert/

├─ electricity.json

├─ magnetic_effects.json

└─ human_eye.json
└─ ...

Tasks

Task 1

Create NCERT JSON files.

Each chapter should contain

- Concepts
- Keywords
- Definitions
- Important Points
- Sample Answers

Example

```
{  
  "chapter": "Electricity",  
  
  "concepts": [  
    {  
      "name": "Electric Current",  
  
      "definition": "Flow of electric charge",  
  
      "keywords": [  
        "charge",  
        "flow",  
        "ampere",  
        "current"  
      ]  
    }  
  ]  
}
```

Task 2

Build textbook loader.

Function

load_chapter()

Returns

Python dictionary

Task 3

Build retriever.

Input

Question

Output

Relevant concept(s)

Task 4

Build rubric generator.

Generate

- Key points
 - Marks distribution
 - Important keywords
-

Task 5

Build evaluation prompt.

Input

Question

Student Answer

Retrieved Concept

Rubric

Output

JSON

```
{  
  "score":8,  
  
  "feedback":"Good explanation but missed Ohm's Law.",  
  
  "missing_points":[...]  
}
```

Task 6

Feedback Formatter

Generate

Strengths

Weaknesses

Missing Concepts

Suggestions

Deliverables

✓ JSON Knowledge Base

✓ Retrieval

✓ Rubric

✓ Evaluation

✓ Feedback

Branch

feature/evaluation-rag